Matthew Braaksma

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RESEARCH INTERESTS

Environment and Resource Economics · Geospatial Analysis · Land Use Economics · Ecosystem Services

EDUCATION

University of Minnesota — Twin Cities, MN
Doctor of Philosophy: Applied Economics — Aug 2022 – Present

San Diego State University — San Diego, CA
Master of Arts: Economics — Aug 2020 – May 2022

Eastern Washington University — Cheney, WA
Bachelor of Arts: Business Administration — Sept 2013 – June 2017
Minor: Economics, Visual Communication Design

SKILLS

Languages and Tools
Proficient: Python (gdal, pandas, geopandas, xarray, rasterio), GDAL, NetCDF, SEALS, InVEST, Quarto, Markdown, R (tidyverse, data.table, terra), Stata
Intermediate: Bash, Git, LaTeX

CURRENT RESEARCH EXPERIENCE

University of Minnesota Climate Adaptation Partnership (MCAP) — Twin Cities, MN
Research Assistant, Supervised by Dr. Heidi Roop — June 2026 – August 2026

- Selected to support MCAP climate research applying geospatial computing to explore processing mobility datasets and analyzing cooling center accessibility under changing temperatures and extreme heat events.
- Prepare and process climate projection datasets (NetCDF, GeoTIFF); develop Python/xarray and reproducible workflows for analysis and visualization.
- Assist with updating online mapping tools (dashboards) and collaborate with Dr. Heidi Roop, Dr. Ryan Harp, and Dena Coffman.

University of Minnesota — Twin Cities, MN
Research Assistant, Supervised by Dr. Justin Johnson — September 2025 – Current

- Lead development of K-SEALS fine-scale Land Use / Land Cover projection framework for Korea, translating Nature Futures scenario narratives into spatially explicit LULC futures.
- Integrated and processed multi-source geospatial layers (land cover, protected areas, wetlands, urban proximity, biodiversity indices, connectivity) to generate per-class hectare demand and spatial allocation inputs.
- Implemented a two-stage pipeline: regional-change (per-class coarse hectare demand) and spatial allocation (fine-scale LULC allocation), producing reproducible GeoTIFF and NetCDF-ready outputs and datasets for downstream analysis.
- Built reproducible Python-based workflows (GDAL, pygeoprocessing, SEALS), automated reporting with Quarto, and delivered analysis-ready outputs for collaborators and modeling partners.

University of Minnesota — Twin Cities, MN
Research Assistant, Supervised by Dr. Justin Johnson — June 2024 – December 2025

- Co-developed "A Framework for Measuring African Ecosystem Services Impacts from Land-Related Policies" with collaborators from the European Commission Joint Research Centre and Wageningen University and Research.

- Contributed to an integrated modeling framework coupling a computable general equilibrium (CGE) model with spatially explicit ecosystem-service models (InVEST, SEALS) to evaluate economic and ecological consequences of land-use policy scenarios in Kenya.
- Implemented data pipelines and analyses using Python and geospatial toolkits (SEALS, InVEST, GDAL), producing reproducible scenario outputs and diagnostics to support policy-relevant assessment.

NatCap TEEMs (The Earth-Economy Modelers) Twin Cities, MN
Student Member August 2022 – Present

- Contribute to collaborative research improving integrated Earth–economy models that combine ecological, climate, and economic data for sustainable development analysis.
- Support efforts to align economic modeling with environmental stewardship through improved data integration and reproducible geospatial workflows.

RESEARCH

Working Papers Dave, Dhaval, Yang Liang, Catherine Maclean, Joseph Sabia, and Matthew Braaksma. "Can Anti-Vaping Policies Curb Drinking Externalities? Evidence from E-Cigarette Taxation and Traffic Fatalities". No. w30670. *National Bureau of Economic Research*, 2022.

Publications McWay, Ryan, and Matthew Braaksma. 2025. "The Political Consequences of Resource Scarcity: Targeted Spending in a Water-Stressed Democracy. A Replication Study of Mahadevan and Shenoy (Journal of Public Economics, 2023)". *World Development Perspectives* 39: 100707. https://doi.org/10.1016/j.wdp.2025.100707

Conference Presentations "Global Spatially Explicit Land Supply Elasticity Estimates for Economic Modeling." 29th Annual Conference on Global Economic Analysis: "Confronting Developmental and Environmental Challenges in Asia and the New Trade Landscape." Global Trade Analysis Project (GTAP), Kyoto, Japan, June 17–19, 2026.

"A Framework for Measuring African Ecosystem Services Impacts from Land-Related Policies." 28th Annual Conference on Global Economic Analysis: "Accelerating Economic Transformation, Resilience, Diversification, and Job Creation." Global Trade Analysis Project (GTAP), Kigali, Rwanda, June 25–27, 2025.

Conference Service Abstract Reviewer, 29th Annual Conference on Global Economic Analysis, Global Trade Analysis Project (GTAP), Kyoto, Japan, June 2026.

TEACHING EXPERIENCE

University of Minnesota Twin Cities, MN
Teaching Assistant Spring 2025
APEC 3002: Managerial Economics Dr. Frances Homans

- Designed and led labs focused on applying economic models to real-world contexts.
- Developed a memo-based assignment on ecosystem services to connect theory with practice.

University of Minnesota Twin Cities, MN
Teaching Assistant, PhD-level Spring 2024
APEC 8213: Econometric Analysis III Dr. Paul Glewwe
APEC 8214: Econometric Analysis IV Dr. Marc Bellemare

- Led R-based labs to help students develop fluency in estimation, diagnostics, and presentation.
- Assisted students in applying econometric methods through problem sets and coding support.

University of Minnesota Twin Cities, MN
Teaching Assistant, PhD-level Fall 2023
APEC 8001: Applied Microeconomic Analysis of Consumer Choice and Consumer Demand Dr. Paul Glewwe
APEC 8002: Applied Microeconomic Analysis of Production and Choice Under Uncertainty Dr. Rodney Smith

- Created recitation slides and guided discussion sections to clarify core theoretical concepts.
- Provided detailed feedback on assignments to support student mastery of technical material.

RECOGNITIONS

Abstract of Distinction
University of Minnesota, CFANS Research Symposium 2026

Teaching Assistant Award for 3000/4000 Level Classes
University of Minnesota, Applied Economics 2024–2025

Teaching Assistant Award for Graduate Classes
University of Minnesota, Applied Economics 2023–2024

Graduate Fellowship
University of Minnesota, Applied Economics 2022

Presidential Graduate Research Fellowship
San Diego State University 2020

McCuen Endowed Economics Scholarship
San Diego State University 2020

Transfer Achievement Scholarship
Eastern Washington University 2013